

Watching Each Other: Partisan Poll-Watchers and the Vote Count

Francisco Pardo

University of Texas at Austin

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Abstract

Party poll-watchers monitor vote counts in many democracies, yet coverage is often uneven. I study three presidential runoffs in Peru using handwritten records from official voting registries, comparing polling stations within schools. A poll-watcher present without a rival shifts the vote margin toward their party by up to 0.4 percentage points; scaled nationally, a party watching half of all polling stations unopposed would move the margin by roughly 0.2 points, the margin by which two of these runoffs were decided. When both parties watch, the effects cancel, null votes fall, and counts take longer, consistent with legitimate contestation of ballots.

JEL Codes: D72, N46

Free and fair elections are foundational to democratic accountability, allowing voters to discipline and replace politicians ([Besley and Case, 1995](#); [Ferraz and Finan, 2008](#)). When citizens cannot trust that their votes are counted as cast, the legitimacy of elected governments suffers ([Berman et al., 2019](#)). Yet concerns about the legitimacy of election

results are widespread: in the most recent round of the World Values Survey, half of the countries had more than a third of respondents indicate that votes are often not counted fairly.¹

This paper studies the people who watch the count. In Peru, as in many US states, parties may place poll-watchers in any polling station (classroom). Poll-watchers observe the count and can contest poll-workers' rulings on individual ballots. Watcher coverage depends on each party's capacity to recruit volunteers, and it is thin: in the three runoffs I study, between 46% and 63% of classrooms counted their votes with no party watcher present. I ask what watchers do to the count when they are present, and what their uneven presence implies for electoral outcomes.

Estimating the effect of poll-watchers on election outcomes is typically challenging because parties recruit more poll-watchers in places where they have more support. Within a school, however, voters are assigned to classrooms alphabetically, so classrooms in the same school serve comparable electorates. I compare classrooms within a school, controlling for each classroom's first-round result and for the presence of the other party's watcher. This identifies the effect of each of the four possible configurations of watcher presence.

I use data from official voting registries scanned and published by Peru's electoral institution (Oficina Nacional de Procesos Electorales, ONPE) for three presidential runoffs: 2011, 2016, and 2021. From these registries, I extract handwritten information about poll-watcher and poll-worker presence, party affiliation, national ID numbers, and names, which I use to construct proxies for age and gender. I also extract timestamps that I use to measure the duration of voting and ballot counting.

I find that a poll-watcher present without a rival shifts the vote margin toward their party by up to 0.4 percentage points, both raising their own party's vote share and

¹See [Figure B.2](#).

lowering their rival's. Scaled to the national level, this effect is large relative to the margins of the 2016 and 2021 elections. A party watching half of all polling stations unopposed would move the national margin by roughly 0.2 percentage points, about the margin by which each election was decided. Importantly, when both are present, the effects cancel and the margin is indistinguishable from classrooms with no poll-watcher.

The evidence suggests that poll-watchers primarily operate through the legitimate contestation of ballots during the count. When both parties' poll-watchers are present, the share of null votes falls and the duration of the count increases by 3.3%, consistent with longer deliberations over disputed ballots. By contrast, I find no significant effects on blank votes or turnout, providing little evidence that fraud or turnout buying drive the results. The effect on the duration of voting is also below half of one percent, consistent with no major interventions while polls are open. The results are robust to alternative samples, including restricting to classrooms that opened on time and to those staffed by the same poll-workers in both rounds.

This paper makes three contributions to the literature. First, it adds to work on partisan poll-watchers,² which has emphasized illegitimate channels, such as removing rival ballots in Argentina (Casas, Díaz and Trindade, 2017) and enforcing vote buying in Mexico (Ascencio and Rueda, 2019), and to related work on discretion over who may vote, exercised by polling-station officials in India (Neggers, 2018) and the United States (Atkeson et al., 2014).³ I document a legitimate channel, the contestation of ballot validity during the count, in a setting where a single official ballot rules out removing rival ballots and a national ID leaves little room for discretion over eligibility. Party watchers can use this channel wherever they observe the count, including in established

²For studies on non-partisan poll-watchers, see Buzin, Brondum and Robertson (2016), Asunka et al. (2019), Hyde (2007), Hyde (2010), Leeffer and Vicente (2019), Sjoberg (2012), Ichino and Schündeln (2012), and Sjoberg (2016).

³This channel operates in contexts where multiple valid identity documents give officials discretion to question eligibility. In Peru, a single national ID leaves little room for such discretion.

democracies such as the United States, which makes the results relevant well beyond settings where more overt manipulation is common.

Second, I study partisan poll-watching in presidential runoffs decided by narrow margins where parties have limited capacity to monitor and influence the count. The 2016 and 2021 Peruvian runoffs were each decided by 0.2 percentage points, and none of the three runoffs in my sample featured an incumbent party. This contrasts with [Casas, Díaz and Trindade \(2017\)](#) and [Ascencio and Rueda \(2019\)](#), who study elections with incumbents and settings where the party with greater coverage had poll-watchers in around 90% of polling stations. With no incumbent in the race, neither side can rely on control of the state, and each must rely on its own organization to monitor the count. This setting allows me to test whether poll-watchers still affect the count when neither party controls the state and both leave many polling stations unwatched. Close races are also more likely to generate challenges to election results: runner-up parties reject outcomes even in elections of high quality, and rejection becomes more common as the margin narrows ([Hernández-Huerta, 2020](#)). In the United States, [Card and Moretti \(2007\)](#) find suggestive evidence that electronic voting had a small effect on the 2000 and 2004 presidential elections, large enough to matter in closely contested states, although they find no evidence that the mechanism was illegal vote manipulation. Narrow-margin elections are common, and electoral systems like that of the United States are particularly exposed to them ([Geruso and Spears, 2026](#)) but they can also follow from how parties mobilize voters ([Levine and Martinelli, 2024](#)).

Third, I introduce a novel measure of the counting process: the duration of ballot counting, constructed from timestamps in official voting registries. This measure provides a direct way to examine whether poll-watchers affect what happens during the count. I find that counting takes 3.3% longer when both parties are present, consistent with more deliberation over disputed ballots. This measure complements existing studies

that typically focus on election outcomes or ballot-level decisions without observing the time required to complete the count.

These results suggest that balanced poll-watcher coverage can improve the process without changing the outcome. When both parties' poll-watchers are present, their effects on the margin cancel and the count instead takes longer, consistent with more deliberation over disputed ballots rather than a change in the result. Symmetric coverage is also the arrangement voters trust most. In survey experiments in the United States, respondents of both parties judge an election fairest when poll-watchers from both parties are present (Cohen and Sheagley, 2024), and in Malawi and Kenya, voters' confidence in tally sheets rises most when they are co-signed by agents from both parties (Mbozi, 2025). Promoting coverage by both parties could therefore strengthen confidence in the count without tilting the result toward either party.

The rest of the paper proceeds as follows. [Section 1](#) describes the Peruvian election system, and [Section 2](#) describes the data used. [Section 3](#) presents the empirical strategy. [Section 4](#) presents the main results and mechanisms, and [Section 5](#) the robustness checks. [Section 6](#) concludes.

1 Elections and Poll-Watchers in Peru

This section describes the institutional features of Peruvian elections relevant to this paper: the electoral rules, how votes are cast, and the role of party poll-watchers. The three elections studied were all closely contested runoffs. The 2011 runoff was won by fewer than half a million votes, a margin of 2.7%. The 2016 and 2021 elections were razor-thin, both decided by fewer than 50,000 votes, corresponding to a margin of 0.2% of votes cast. Notably, the same party lost all three elections, while the winning party

was different in each.⁴

1.1 Election Rules

In Peru, voting is compulsory,⁵ and the presidency is generally decided in two rounds. In the first-round, voters choose among multiple presidential candidates: 11 parties competed in 2011, 14 in 2016, and 18 in 2021. If no candidate receives more than 50% of the vote, the two leading candidates advance to a runoff. The second-round determines only the presidency, as congress has already been elected by that point.⁶ Presidential reelection is not permitted in Peru, and given the high degree of party fragmentation, no incumbent party participated in any of the three runoffs studied in this paper.

1.2 How Votes are Cast: Schools and classrooms

Elections are held at electoral sites, which in Peru are typically schools. Each voter must vote within their district of residence, but may select their school ahead of the election, typically choosing one near their home.⁷ Within each school, voters are assigned to a polling station (a classroom) by the electoral institution (ONPE), in alphabetical order. Each classroom generally serves between 150 and 300 voters and is presided over by one poll-worker, assisted by two others, all drawn from the registered voters of that classroom and assigned by ONPE ahead of the election. These poll-workers are responsible for setting up and reviewing election materials, verifying voter IDs, and

⁴Fuerza Popular went on to win the 2026 runoff by 0.26 percentage points. The 2026 election also shows how logistical failures can feed fraud narratives, particularly in the first-round, when partisan poll-watchers are almost absent. The late arrival of voting materials delayed the opening of many classrooms in Lima, voting was extended by a day in some locations, and the head of ONPE resigned before the runoff. Some parties alleged fraud, but the OAS and European Union observation missions found no evidence of it.

⁵Fines for not voting are between 5 USD and 25 USD

⁶Because there are no strong parties in Peru, votes are highly disperse and hence, it is often the case that no candidate has won a majority in congress before heading to the second-round

⁷Around 10% of voters choose their own school; the rest are assigned to a school within their district by the electoral institution.

conducting the count. Additionally, parties may assign poll-watchers to supervise this process and raise concerns or contest decisions during the vote count. Upon entering the room, a voter approaches the poll-workers, presents their ID, and receives a ballot listing all candidates. The voter then casts their ballot behind a private booth and deposits it in a ballot box.

1.3 Party Poll-Watchers

Each party may assign poll-watchers to any classroom to monitor the voting process and the vote count at the end of the day. Because parties generally do not have enough poll-watchers to cover all classrooms, they must allocate them selectively, leaving some classrooms unmonitored. My identification strategy relies on this allocation being conditionally random within a school, which I discuss in [Section 3](#). During the vote, poll-watchers are seated alongside the poll-workers inside the classroom. [Figure B.1](#) illustrates this setup, showing the exterior of a school, voters lined up inside a classroom, and the key actors present during the vote: poll-workers, a voter at the booth, and party poll-watchers.

Poll-watchers have an official role before, during, and after the election. Before voting begins, they verify that the voting booth is properly set up and that the classroom has all the necessary ballots. During voting, they may verify that the voting booth has no marks and raise observations if irregularities arise. After voting closes, they ensure that the total number of ballots matches the total number of voters, and may contest poll-workers' rulings on individual ballots.

If poll-watchers affect the count, their presence should raise their own party's vote share and lower their rival's, widening the margin between the two. The main way they can do this is legitimate: contesting poll-workers' rulings on whether individual

ballots are valid.⁸ Voters make a range of mistakes when marking their ballots (Dee, 2007). In Peru these mistakes surface as ballots whose validity the poll-workers must judge. During the count, a poll-watcher may argue that a ballot cast for their party was wrongly deemed null, for instance when a voter marked the candidate's picture instead of the party logo. If the poll-workers agree, the vote counts and the party's share rises. Conversely, a poll-watcher may argue that a ballot counted for the rival should be invalidated, for instance when a voter circled rather than crossed their choice or wrote on the ballot, which lowers the rival's share. Poll-workers may be more receptive to arguments for counting a vote than for discarding one, since the latter can look like suppressing a voter's choice. If so, poll-watchers should have a larger effect on their own party's share than on their rival's. Two further implications follow. When both parties' poll-watchers are present, both vote shares could rise as fewer ballots are deemed null. And because each dispute takes time, counts should take longer when more ballots are contested, especially when both parties' poll-watchers are present to argue over the same ballots.

Prior work has also documented illegitimate channels, but Peru's institutions make most of them unlikely here. In Argentina, each party prints and distributes its own ballots, and Casas, Díaz and Trindade (2017) argue that poll-watchers affect results mainly by removing opposition ballots; in Peru, a single official ballot lists all candidates. In India, poll-workers exercise discretion over voter eligibility in areas with low identity document coverage, which Neggers (2018) finds to be a source of electoral bias; in Peru, voters must present a national ID, which leaves little room for such discretion. In Mexico, Ascencio and Rueda (2019) find that reading voters' names aloud as they approach the classroom could be used by poll-watchers as a tool to irregularly control voting behavior by facilitating turnout buying enforcement; In Peru, poll-workers are

⁸It is unclear whether poll-watchers are correcting errors or introducing a pro-party bias that deviates from the intended vote.

not instructed to read names out loud. However, poll-watchers could still intimidate voters or otherwise disrupt voting while polls are open, which would lower turnout or lengthen the voting process. Finally, poll-watchers could facilitate or deter outright fraud, such as filling in blank ballots or those of absent voters, which would appear as higher turnout or fewer blank votes. I test for these traces in [Section 4](#).

2 Data

My data come from two official sources from ONPE, the Peruvian electoral institution.⁹ The first source is the official vote counts, which ONPE publishes publicly at the classroom level for both rounds of each election. These provide vote totals by party, blank votes, null votes, and turnout for each of the classrooms. The amount of schools, classrooms per school, and voters per classroom varies by year as shown in the first panel of [Table 1](#). The second source is ONPE’s official voting registries, one per classroom, scanned and uploaded after each election. After the count is completed, the three poll-workers destroy the ballots and record the total votes received by each party on the registry. Any poll-watchers present also sign the sheet and write down their party affiliation.

This handwritten information is my primary source of data on poll-worker and poll-watcher presence and party affiliation. In addition, both poll-workers and poll-watchers are required to write down their national ID number and full name, which I use to construct proxies for age and gender. From these records, I also extract timestamps that log the opening and closing times of both the voting period and the vote count, which I use to construct the duration of each for every classroom. Further details on

⁹I have only considered votes by citizens within Peru, excluding those cast in a foreign country. This is due to difficulties in obtaining some of the information from foreign registries due to the differences in format.

the data collection and processing procedure are provided in [Appendix A](#).

[Table 1](#) reports descriptive statistics for all three elections studied in this paper, which were decided by narrow margins. The 2011 runoff was won by fewer than half a million votes, a margin of 2.7%. The 2016 and 2021 elections were razor-thin: both were decided by fewer than 50,000 votes, corresponding to margins of just 0.2% of votes cast. With narrow margins like these, even small differences in how votes are counted could plausibly determine the outcome.

Poll-watcher coverage is considerably lower than in comparable settings. Across the three elections, between 37% and 54% of classrooms had any poll-watcher present, meaning that in a large share of classrooms the vote count was conducted with no party supervision at all.¹⁰ Additionally, when poll-watchers are present, they tend to cluster together rather than monitor classrooms independently. Both poll-watchers are present in around a third of classrooms, much more frequently than either party alone, suggesting that parties respond to each other’s assignments. As I will show in the next section, the identification strategy relies on poll-watcher allocation being unrelated to potential outcomes within a school, after controlling for predicted second-round vote shares based on first-round results and the presence of the opposing party’s poll-watcher.

3 Empirical Strategy

The main empirical challenge is that poll-watcher assignment is not random across the country. Because poll-watchers are volunteers, parties find it easier to recruit them in areas where they have more support. Panel (a) of [Figure 1](#) plots poll-watcher presence against party support at the district level and shows this relationship clearly. A simple

¹⁰This stands in contrast to settings like Argentina ([Casas, Díaz and Trindade, 2017](#)) and Mexico ([Ascencio and Rueda, 2019](#)), where coverage is substantially higher. In both countries, the party with greater coverage had poll-watchers present in around 90% of classrooms.

comparison across schools would therefore confound the effect of poll-watchers with differences in underlying electoral preferences.

Within a school, however, voters are assigned to classrooms in alphabetical order, which produces groups of voters with similar ideological preferences across classrooms within the same school. Differences in poll-watcher presence across these classrooms therefore provide a source of within-school variation that is less likely to reflect differences in electoral preferences.¹¹ I use this variation to estimate the effect of poll-watchers on the vote margin using the following specification:

$$VoteMargin_{sc} = \alpha_s + \delta_W Obs_{sc}^W + \delta_L Obs_{sc}^L + \delta_{both} Obs_{sc}^{both} + \beta X_{sc} + \epsilon_{sc} \quad (1)$$

Where $VoteMargin_{sc}$ ¹² is the vote share margin between the two parties in classroom c of school s . I label the eventual runoff winner as W and the runner-up as L throughout. Obs_{sc}^W , Obs_{sc}^L , and Obs_{sc}^{both} are indicators for whether a poll-watcher from the winning party alone, the losing party alone, or both parties are present in the classroom. α_s denotes school fixed effects and X_{sc} includes controls for other variables that may be related to poll-watcher assignment, discussed below. I expect δ_W to be positive, δ_L to be negative, and δ_{both} to be smaller in magnitude than either, as the two poll-watchers would offset each other to some extent.

The key assumption is that poll-watcher presence is unrelated to potential outcomes within a school. From conversations with electoral officials and party coordinators during election day, it appears that poll-watcher assignment does not follow any sophisticated or nationally coordinated strategy at the within-school level.

¹¹Cantú (2014) and Casas, Díaz and Trindade (2017) find evidence consistent with this in the Mexican and Argentinian contexts respectively

¹²Defined as the winner's vote share minus the loser's vote share in percentage points

Below, I discuss potential threats to identification and how I address them. First, alphabetical assignment does not guarantee that classrooms within a school are perfectly homogeneous. Parties could observe differences across classrooms that, if correlated with poll-watcher assignment, would introduce bias. [Figure B.5](#) plots vote shares and poll-watcher presence by the first initial of the lead poll-worker’s surname and shows no systematic variation across initials, consistent with alphabetical assignment producing groups of voters with similar underlying preferences. However, because classrooms have the same group of voters in both the first and second-round, parties could in principle use first-round results to inform their poll-watcher assignments within a school. Panel (b) of [Figure 1](#) shows that even within a school, first-round vote shares are a strong predictor of second-round vote shares. If parties exploited this and assigned their poll-watchers to classrooms where they expected to perform better, estimates would be biased away from zero.

I address this by controlling for the predicted second-round vote margin, $\widehat{VoteMargin}_{sc}$, in all specifications. For each runoff candidate $p \in \{W, L\}$, I regress the candidate’s second-round vote share on the first-round vote shares of all parties, with school fixed effects, separately for each election:

$$VoteShare_{sc}^p = \mu_s^p + \sum_{k=1}^K \gamma_k^p FirstRound_{sc}^k + u_{sc}^p, \quad p \in \{W, L\}, \quad (2)$$

where $FirstRound_{sc}^k$ is the first-round vote share of party k in classroom c of school s . The predicted margin is the difference between the two fitted values, including the estimated school fixed effects: $\widehat{VoteMargin}_{sc} = \widehat{VoteShare}_{sc}^W - \widehat{VoteShare}_{sc}^L$. Because it is built from each classroom’s own first-round results, this control captures the differences across classrooms within a school that show up in first-round voting, which are the ones parties could most plausibly observe and act on. As a further check, I use the

predicted margin as a placebo outcome, estimating equation (1) with $\widehat{VoteMargin}_{sc}$ on the left-hand side and omitting it from the controls. [Section 5.3](#) reports these estimates.

Second, poll-watchers can observe the presence of the other party’s poll-watcher on election day and may respond to it. Panels (c) and (d) of [Figure 1](#) show this clearly: panel (c) plots the observed share of classrooms with both parties’ poll-watchers against the share expected under independent random assignment, and panel (d) compares the actual distribution to both the random and maximum poll-watcher match benchmarks. Both panels indicate that poll-watchers from both parties are considerably more likely to be present in the same classroom than would be expected under independent random assignment. If both parties assigned poll-watchers independently of each other, one would expect both to be present in around 23% of classrooms. In practice, both are present in 34% of classrooms, close to the theoretical maximum of 35% if parties always matched their opponent’s poll-watcher presence, indicating that parties systematically place poll-watchers where the other party has already done so. This pattern matches [Ascencio and Rueda \(2019\)](#), who estimate a model of party representation in Mexico and find that parties send representatives where they expect their opponents to send their own, which they read as evidence of a protective motive. An important advantage of the Peruvian setting is that there are only two parties in the runoff, which means I can fully account for all four possible combinations of poll-watcher presence: no poll-watcher (omitted), winner party alone, loser party alone, and both. This is less feasible in settings with more parties, such as [Casas, Díaz and Trindade \(2017\)](#) and [Ascencio and Rueda \(2019\)](#), where accounting for all possible interactions is impractical. By including all four cases explicitly in the specification, I control for this form of strategic co-location.

Third, a potential source of measurement error in poll-watcher presence arises if party poll-watchers stepped in to replace absent poll-workers on election day. In this case, they

would appear in the voting registries as poll-workers rather than poll-watchers, leading their presence to be undercounted. Because a poll-watcher acting as a poll-worker has more influence than one sitting alongside the poll-workers in their designated role, this would likely bias my estimates toward zero. I assess this by restricting the sample to classrooms whose poll-workers were the same in both the first and second-round. Since poll-watchers are not present in the first-round, any replacement would not affect first-round poll-worker composition. [Section 5.1](#) reports these estimates. A related concern is that poll-workers themselves may have had undisclosed party affiliations, and that parties assigned poll-watchers accordingly. This is unlikely to be a major issue given that formal party affiliation is very low in Peru, at under 0.5% of the population according to the 2021 household survey, and poll-workers are drawn at random from the registered voters of each classroom.

4 Results

4.1 The Effect of Poll-Watchers on the Vote Margin

[Table 2](#) reports results from various specifications of equation (1). Columns (1) through (5) progressively add fixed effects and controls on a common sample, while column (6) reports the preferred specification on the full sample. In all specifications with school fixed effects, a party’s poll-watcher present alone shifts the vote margin in that party’s favor. In column (6), the margin moves 0.31 percentage points toward the winner when only the winner’s poll-watcher is present, and 0.38 points toward the loser when only the loser’s is.

The first specification in column (1) regresses the vote margin on each poll-watcher presence category. Column (2) adds district fixed effects, which captures most of the

variation and causes the largest drop in coefficients from 10 to around 1 percentage point. Column (3) adds school fixed effects, which causes the coefficients to drop further to 0.48 percentage points for the winner party poll-watcher, and to -0.31 percentage points for the loser party. This illustrates the selection problem when relying on across-school variation in poll-watcher presence: parties tend to have greater poll-watcher presence in schools where their support is higher.

Importantly, once school fixed effects are included and identification relies on within-school variation in poll-watcher presence, the coefficients remain stable as first-round controls and poll-worker demographics¹³ are progressively added in columns (4) and (5). Since demographic controls are not always available and their inclusion does not affect the estimates, column (6) reports results for the full sample excluding the demographic controls from column (5). This serves as the preferred specification throughout the rest of the paper.

The stability of the estimates to the addition of first-round results and salient characteristics of classroom poll-workers indicates that the results are not driven by these observable sources of strategic assignment. It cannot, however, rule out selection on unobserved classroom characteristics, which I examine further in [Section 5](#). Furthermore, as discussed in [Section 3](#), panels (c) and (d) of [Figure 1](#) show that one factor informing poll-watcher assignment is the presence of poll-watchers from the opposing party. Since the runoff election involves only two parties and all four poll-watcher configurations are observed, I account for this strategic response by estimating separate effects for each configuration.

These estimates come from a setting with less room for manipulation than prior work has studied. [Casas, Díaz and Trindade \(2017\)](#) find that observers raise their party’s vote count by 1.5% on average in Argentina, and by up to 6% for some parties,

¹³indicators for gender and age, proxied from first names and national ID numbers

where each party prints and distributes its own ballots. In Peru, a single official ballot rules out removing rival ballots. Even with these constraints, I show that contesting ballot validity is by itself enough to move the count.

4.2 Mechanisms

Section 1, described one legitimate channel, the contestation of ballot validity, and several illegitimate ones. Columns (2) through (8) of Table 3 test the traces each would leave.

Starting with the vote share decomposition in columns (2) and (3), the results are consistent with poll-watchers increasing their own party's vote share and reducing their rival's. When the winning party's poll-watcher is present alone, the winner's vote share increases by 0.16 percentage points and the loser's falls by a similar amount. The pattern is symmetric for the losing party's poll-watcher. However, in this case the effect on their vote share is somewhat larger than the effect on their rival's, which is consistent with poll-watchers being more effective at defending their own votes than at contesting their opponent's, as discussed in Section 1.

The evidence points toward the legitimate channel as the primary mechanism. When both poll-watchers are present, the vote margin is not significant, but null votes fall by 0.08 percentage points. Importantly, this is reinforced by the fact that count duration increases by 1.3 minutes, representing a 3.3% increase over the mean of 38.3 minutes, consistent with longer deliberations over disputed ballots during the count. By contrast, the effects on blank votes and turnout are small and generally insignificant across all configurations, which argues against fraud or turnout buying as relevant mechanisms. The effects on voting duration are also small relative to the mean of 537.1 minutes, suggesting that poll-watcher presence does not meaningfully alter the voting process itself.

5 Robustness Checks

Results from the main specification are robust to a number of alternative samples and specifications, all of which are summarized in [Figure 2](#).

5.1 Classrooms with the Same Poll-Workers as the First-Round

One concern is that poll-watchers may have replaced absent poll-workers on election day, appearing in the voting registries as poll-workers rather than poll-watchers. To address this, I identify these classrooms from the poll-worker IDs recorded in the voting registries. Matching IDs across rounds is imprecise and tends to produce false negatives rather than false positives, so this restricted sample is best understood as a conservative subset of classrooms where replacement is unlikely to have occurred. Results are similar to the main estimates, suggesting that unobserved poll-watcher replacement is not widespread enough to meaningfully affect the findings.

5.2 On-Time Classrooms

A related proxy for poll-worker replacement is whether a classroom opened on time. When all poll-workers show up as assigned, the classroom generally opens on schedule; delays are more likely when a poll-worker is absent and needs to be replaced. I therefore restrict the sample to classrooms that opened on time, using the opening timestamps recorded in the voting registries. Results are again similar to the main estimates, further supporting the conclusion that unobserved poll-watcher replacement is not driving the findings.

5.3 Placebo Test: Predicted Second-Round Votes

Figure 2 also displays, as a placebo test, the effect of each poll-watcher configuration on the predicted second-round vote margin based on first-round results. These effects are not significantly different from zero across most configurations, supporting the identifying assumption that parties are not systematically assigning poll-watchers to classrooms where they expect to perform better. In some cases the placebo estimates are slightly positive, suggesting a mild tendency for parties to favor classrooms where the winning party had a projected advantage. However, the loser party coefficient goes in the opposite direction, and any remaining tendency is in any case addressed by controlling for predicted second-round vote shares in all specifications.

6 Conclusion

I find that a poll-watcher present without a rival shifts the vote margin toward their party by up to 0.4 percentage points, raising their own party's vote share and lowering their rival's. When both parties' poll-watchers are present, their effects offset and the margin is indistinguishable from classrooms with no poll-watcher. Because the two parties largely match each other's placements, one-sided coverage is uncommon in the elections I study. If coverage resembled that of other countries from prior studies, one side alone would watch roughly half of all stations, enough to move the national margin by about 0.2 percentage points, close to the margins that decided the 2016 and 2021 runoffs.

The evidence points toward the legitimate contestation of ballots during the count as the primary mechanism. Poll-watchers appear to be more effective at defending their own party's votes than at contesting their opponent's votes, consistent with poll-workers

being more receptive to arguments in favor of counting a vote than to arguments for invalidating one. This is corroborated by the effect on count duration: when both poll-watchers are present, the duration of the ballot count increases by 3.3%, consistent with longer deliberations over disputed ballots, while the effects on blank votes, turnout, and voting duration are small and insignificant across all configurations.

These findings are relevant beyond the Peruvian context. Narrow-margin elections are common, and electoral systems like that of the United States are particularly exposed to them ([Geruso and Spears, 2026](#)) and they can also follow from how parties mobilize voters ([Levine and Martinelli, 2024](#)). Balanced coverage neutralizes the effect on the margin and is also the arrangement voters trust most ([Cohen and Sheagley, 2024](#); [Mbozi, 2025](#)), which makes it a practical objective for election administrators.

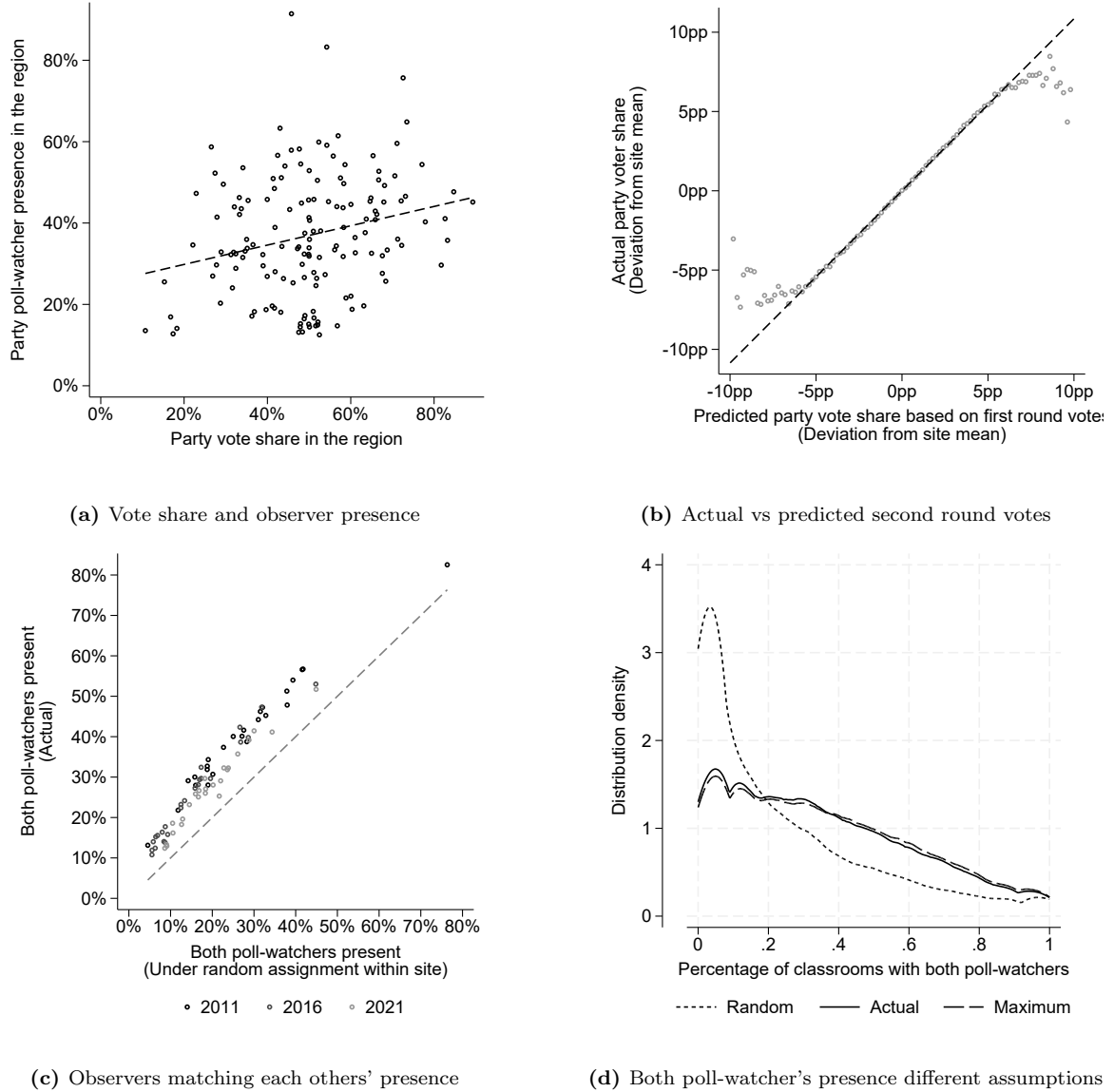
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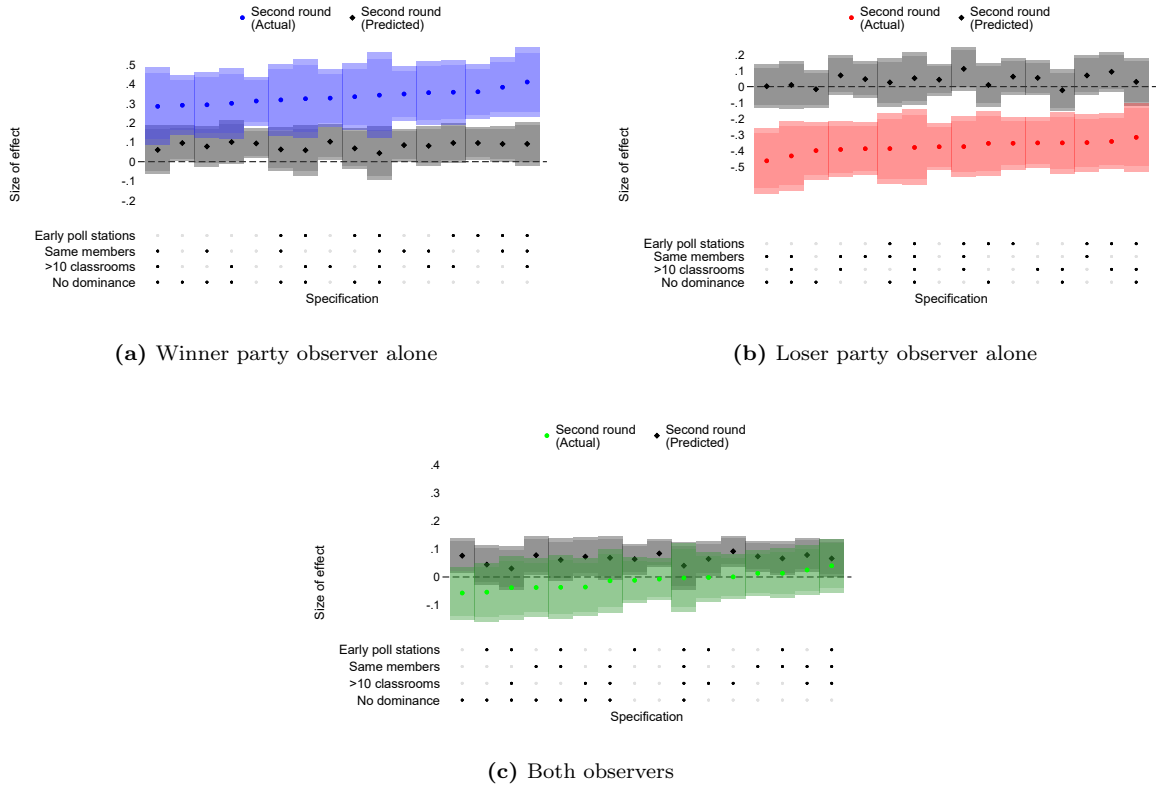
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Figure 1: Identification strategy assumptions



Notes: Panel (a) shows the relationship between party support and poll-watcher presence. Panel (b) shows that even within a polling site, first round votes are a strong predictor of second round votes. Panel (c) plots, for each region-year, the share of polling stations with poll-watchers from both parties on the y-axis against the share that would be expected under random within-site assignment on the x-axis. Panel (d) shows the distribution of classrooms with poll-watchers from both parties under three assignment strategies: (i) random assignment by each party independently (short dash), (ii) the actual observed assignment (solid line), and (iii) the maximum possible overlap, if each party perfectly matched the other's poll-watcher placement (long dash). Panels (c) and (d) together show that poll-watchers from both parties are considerably more likely to be present in the same classroom than would be expected under independent random assignment.

Figure 2: Effect of poll-Watchers



Notes: I show effects on the vote share margin (winner minus loser) of different observer assignments: winner party alone in panel (a), shown in blue; loser party alone in panel (b), shown in red; and both observers in panel (c), shown in green. In each case I show results on the predicted vote share margin based on first round results in grey. I do this for different samples: tables that open on time, with more than 10 classrooms, with the same set of table members than in the first round and without areas with strong domination of one party. 90% and 95% intervals are displayed.

Table 1: Descriptive statistics

	2011	2016	2021
<i>Panel A: Electoral Setting</i>			
Districts	2,176	1,960	2,072
Polling Sites (schools)	4,924	5,234	11,841
Polling-stations (classrooms)	107,436	77,290	86,482
Classrooms per school	21.8	14.8	7.3
Registered voters per classroom	185.7	296.5	292.3
<i>Panel B: First Round Results</i>			
<i>Participation</i>			
Votes per polling-station	155.5	243.9	207.6
Turnout (%)	84.0	82.1	71.1
<i>Vote (%)</i>			
Winner party	27.8	17.3	15.4
Loser party	20.7	32.7	10.9
Other parties	39.3	32.6	55.0
Blank	8.8	11.9	12.4
Null	3.4	5.5	6.3
<i>Panel C: Second Round (Runoff) Results</i>			
<i>Participation</i>			
Votes per polling-station	153.3	237.3	218.0
Turnout (%)	82.9	80.4	75.1
<i>Vote (%)</i>			
Vote margin	2.7	0.2	0.2
Winner party	48.2	46.9	46.9
Loser party	45.5	46.6	46.6
Blank	0.7	0.8	0.6
Null	5.6	5.7	5.9
<i>Poll-watcher presence</i>			
None	46.4	63.3	52.7
Winner party alone	3.8	5.6	5.1
Loser party alone	8.4	2.8	12.9
Both	41.4	28.2	29.3
<i>Duration (minutes)</i>			
Vote duration	470.6	467.0	700.3
Vote count duration	33.5	40.9	40.7

Note: This table shows descriptive statistics for the first and second (runoff) rounds of the three presidential elections studied in this paper (2011, 2016, and 2021). Panel A shows the aggregate and average characteristics of the electoral setting. Panel B shows results for the first round. In none of the three elections did any party obtain more than 50% of the votes, which triggered a runoff election. Winner and Loser parties refer to the eventual winner and runner-up of the second round. Turnout is computed as the share of registered voters who cast a ballot. Vote shares are expressed as a percentage of total votes cast, including blank and null votes. Panel C shows results for the second round. The vote margin is defined as the difference in vote share between the winning and losing party, with margins of approximately 0.2% in the 2016 and 2021 elections. Pollwatcher presence categories are mutually exclusive and reported as a percentage of classrooms. Vote duration refers to the total number of minutes the polling station was open for voting. Vote count duration refers to the number of minutes taken to count the ballots after closing

Table 2: Effect of poll-watchers on the vote margin

	Vote margin in percentage points (winner - loser)					
	(1)	(2)	(3)	(4)	(5)	(6)
Winner only	10.75*** (0.57)	0.95*** (0.12)	0.48*** (0.09)	0.41*** (0.08)	0.41*** (0.08)	0.31*** (0.06)
Loser only	-18.67*** (0.79)	-1.11*** (0.17)	-0.31*** (0.11)	-0.38*** (0.09)	-0.38*** (0.09)	-0.38*** (0.07)
Both	-3.21*** (0.35)	0.22*** (0.08)	0.10* (0.06)	-0.00 (0.05)	-0.01 (0.05)	-0.01 (0.04)
School FE	No	No	Yes	Yes	Yes	Yes
District FE	No	Yes	No	No	No	No
First round controls	No	No	No	Yes	Yes	Yes
Poll-workers demographics	No	No	No	No	Yes	No
Mean	1.39	1.39	1.39	1.39	1.39	2.14
N	170,412	170,412	170,412	170,412	170,412	258,288

Note: This table reports estimates of the effect of poll-watcher presence on the second round vote margin, expressed in percentage points as the difference in vote shares between the winner and runner-up. The three poll-watcher presence categories are mutually exclusive: *Winner only* indicates that only the winning party had a poll-watcher present, *Loser only* indicates that only the losing party had a poll-watcher present, and *Both* indicates that both parties had a poll-watcher present. The omitted category is polling stations with no poll-watcher present. Columns (1) through (5) report results for a common subsample with complete observations for all variables, for comparability. Column (1) reports the baseline estimate with no controls or fixed effects. Column (2) adds district fixed effects. Column (3) adds polling-site (school) fixed effects, which captures most of the variation in poll-watcher assignment and its relationship with vote margins. Column (4) further adds a predicted vote margin based on first round results (see equation (2)). Column (5) adds poll worker demographics, including a gender indicator proxied by first name and an above-30 age indicator proxied by national ID number. Column (6) shows the preferred specification, which uses the full sample and excludes the demographic controls from column (5). Standard errors are clustered at the polling-site level and reported in parentheses. ***p<0.01, **p<0.05, *p<0.1.

Table 3: Effect of poll-watchers on vote shares, turnout, vote duration, and count duration

	Vote shares (percentage points)					Participation, and voting and count duration		
	(1) Margin (Winner – Loser)	(2) Winner	(3) Loser	(4) Null	(5) Blank	(6) Turnout	(7) Voting duration (min)	(8) Count duration (min)
Winner only	0.31*** (0.06)	0.16*** (0.04)	-0.16*** (0.04)	0.01 (0.06)	-0.01** (0.01)	-0.00 (0.03)	-0.21 (0.38)	0.13 (0.22)
Loser only	-0.38*** (0.07)	-0.12** (0.05)	0.23*** (0.05)	-0.11 (0.07)	0.00 (0.01)	-0.02 (0.04)	-0.90* (0.49)	-0.56* (0.30)
Both	-0.01 (0.04)	0.05* (0.03)	0.03 (0.03)	-0.08** (0.04)	0.00 (0.00)	0.00 (0.02)	-0.41* (0.24)	1.27*** (0.15)
School FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
First round controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean	2.14	47.82	45.68	5.76	0.74	80.79	537.09	38.26
N	258,288	258,288	258,288	258,288	258,288	258,289	199,342	206,570

Note: This table reports estimates of the effect of poll-watcher presence on second round outcomes at the polling station level. The three poll-watcher categories are mutually exclusive: *Winner only* indicates that only the winning party had a poll-watcher present, *Loser only* indicates that only the losing party had a poll-watcher present, and *Both* indicates that both parties had a poll-watcher present. The omitted category is polling stations with no poll-watcher present. Columns (1) through (5) report effects on vote shares, expressed as a percentage of total votes cast. Column (1) reports the effect on the margin between the winner and loser. Columns (2) and (3) report effects on the winner and loser vote shares respectively. Columns (4) and (5) report effects on null and blank vote shares. Column (6) reports the effect on turnout, measured as the share of registered voters who cast a ballot. All share variables are expressed in percentage points (0 to 100). Columns (7) and (8) report effects on voting duration and count duration in minutes respectively. The smaller sample in these columns reflects missing duration data for some polling stations due to difficulties extracting this information from the voting registries. All specifications include polling-site fixed effects and first round controls, defined as the predicted second round vote share based on first round results (see equation (2)). Standard errors are clustered at the polling-site level and reported in parentheses. ***p<0.01, **p<0.05, *p<0.1.

Appendix

A1 Data Collection

To construct the poll-watcher presence data, I used a combination of direct data access, image-to-text software, manual review, and AI image recognition. The steps followed were: (i) accessing vote counts from the first and second rounds directly through the ONPE database, along with the 271,208 scanned voting registry records containing poll-watcher information; (ii) using Layout-Parser ([Shen et al. \(2021\)](#)), an image-to-text software, to analyze the handwritten sections of each registry and assign poll-watcher status to each classroom, this software identifies text blocks as shown in [Figure A.1](#), but because it is designed for printed rather than handwritten text, recognition is imperfect; (iii) since only one poll-watcher per party is permitted, cases where two poll-watcher text blocks are detected are coded as having both parties present, even when party affiliation cannot be independently confirmed for each; (iv) once a preliminary assignment to each poll-watcher presence category is determined, I proceed to manually review the registries to correct any misclassifications;¹⁴ and (v) all registries are then processed through AI image recognition¹⁵ to more precisely identify timestamps¹⁶, national ID numbers, and full names of poll-workers and poll-watchers, as well as the

¹⁴The manual review was conducted separately for each pre-assigned group, which expedited the process since most entries were already correct.

¹⁵Cropped images for each timestamp and each poll-worker and poll-watcher entry were analyzed using Gemini 2.5 Flash Lite.

¹⁶To validate the accuracy of the extracted timestamps, I manually reviewed a random sample of 30 classrooms per year for each of the four time variables (classroom opening, closing, and vote count start and end times). Overall, 5.5% of extracted timestamps do not match the manually verified values. Error rates are substantially higher in 2011 (12.1%) than in 2016 (2.1%) and 2021 (2.1%), likely reflecting lower document quality in the earlier election. A large share of errors in 2011 correspond to cases where the extraction algorithm returned a missing value despite a legible timestamp being present in the image. Restricting the sample to observations where the algorithm did return a value, the error rate falls to 2.4% overall, 5.5% in 2011, 0.4% in 2016, and 1.3% in 2021, indicating that conditional on extraction, the algorithm reads timestamps accurately. These error rates are small and unlikely to meaningfully affect my estimates.

party affiliation of poll-workers, which is also used to validate and further improve poll-watcher identification.

Figure A.1: Voting registry and algorithm to detect handwritten text

ACTA ELECTORAL → TOTAL DE ELECTORES HABILITADOS: 300

MESA DE SUFRAGIO N° **036267**

DEPARTAMENTO: **LIMA** PROVINCIA: **LIMA** DISTRITO: **CARABAYLLO**

C ACTA DE ESCRUTINIO

Siendo las **4:06 p.m.** del 06 de junio de 2021, se inició el ACTO DE ESCRUTINIO.

MUY IMPORTANTE: Escriba con números legibles como estos: 9 1 2 3 4 5 6 7 8 9

ORGANIZACIONES POLITICAS		TOTAL DE VOTOS
1	PARTIDO POLITICO NACIONAL PERU LIBRE	80
2	FUERZA POPULAR	132
VOTOS EN BLANCO		1
VOTOS NULOS		12
VOTOS IMPUGNADOS		
TOTAL DE VOTOS EMITIDOS →		225

OBSERVACIONES →

Siendo las **4:48 p.m.** finalizó el ACTO DE ESCRUTINIO.

FIRMA, NOMBRES, APELLIDOS Y DNI DE MIEMBROS DE MESA (OBLIGATORIO)

FIRMA Y DNI DE LOS PERSONEROS

Notes: The figure on the left shows how the Layout-Parser software identifies textblocks from the print and handwritten information in the right. Personal information from poll-workers and poll-watchers has been blurred for this example.

A2 Other Figures and Tables

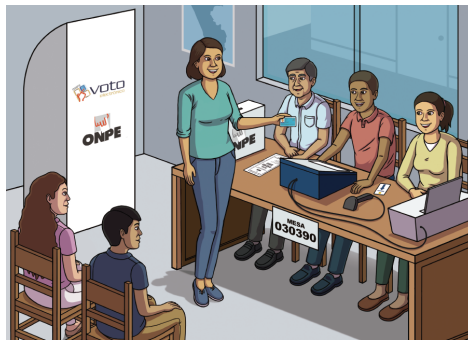
Figure B.1: Election day in Peru



(a) Electoral site (school)



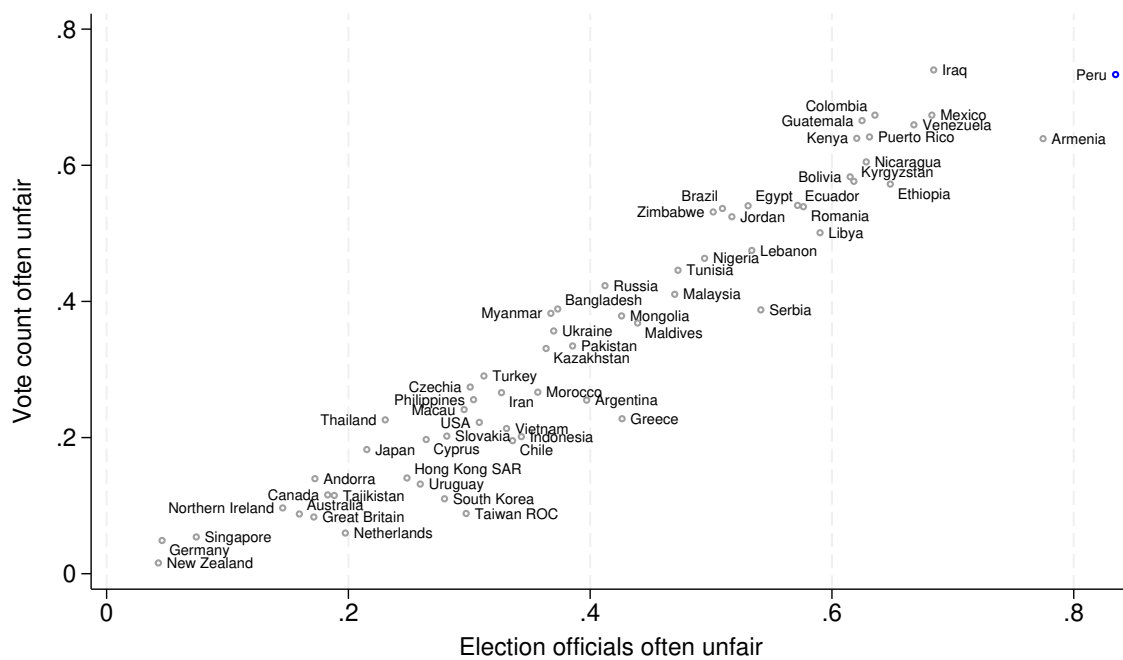
(b) Line outside electoral table (classroom)



(c) Table members (right), voter (center) and observers (left) in a voting table (classroom)

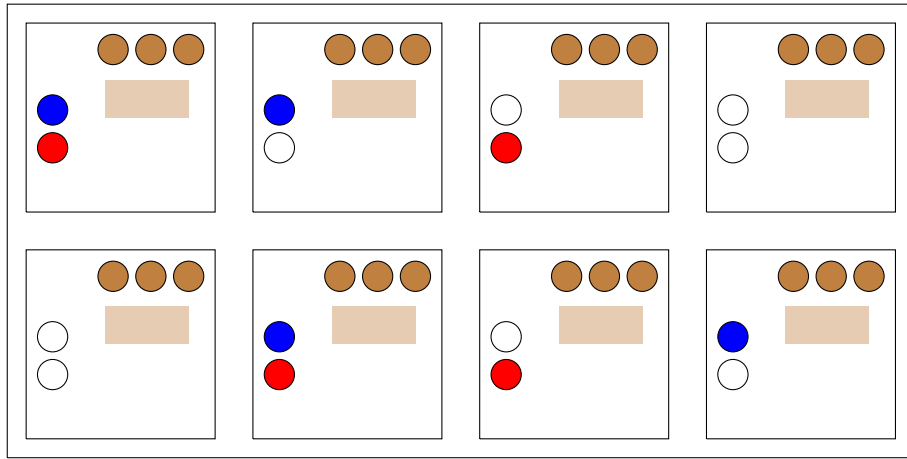
Notes: These images show how voting takes place on election day. Image (a) shows the outside of a polling site (school), where voters can see which classroom they are assigned to based on their ID and last name. Image (b) shows voters in a polling station (classroom) lined up and waiting for their turn to vote. Image (c) shows the different actors inside the classroom: three poll-workers (top right), drawn from the voter pool assigned to that classroom, are responsible for overseeing the voting process, verifying IDs, and counting votes. The voter (middle) steps behind a voting booth to cast their ballot. The party poll-watchers (bottom left) monitor the process and may flag any irregularities. During the vote count, they may challenge any ballot they believe was counted improperly.

Figure B.2: Percentage of individuals who consider election officials and vote count to be often unfair



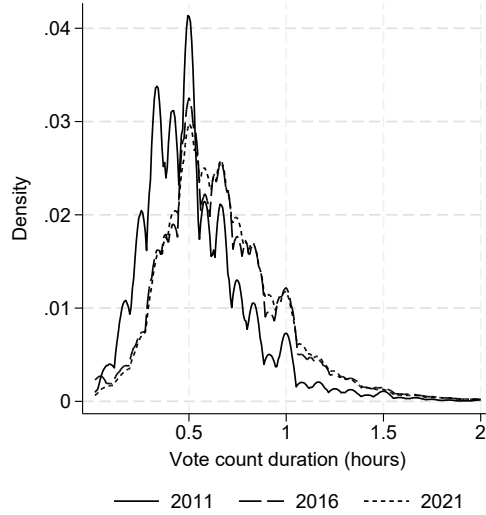
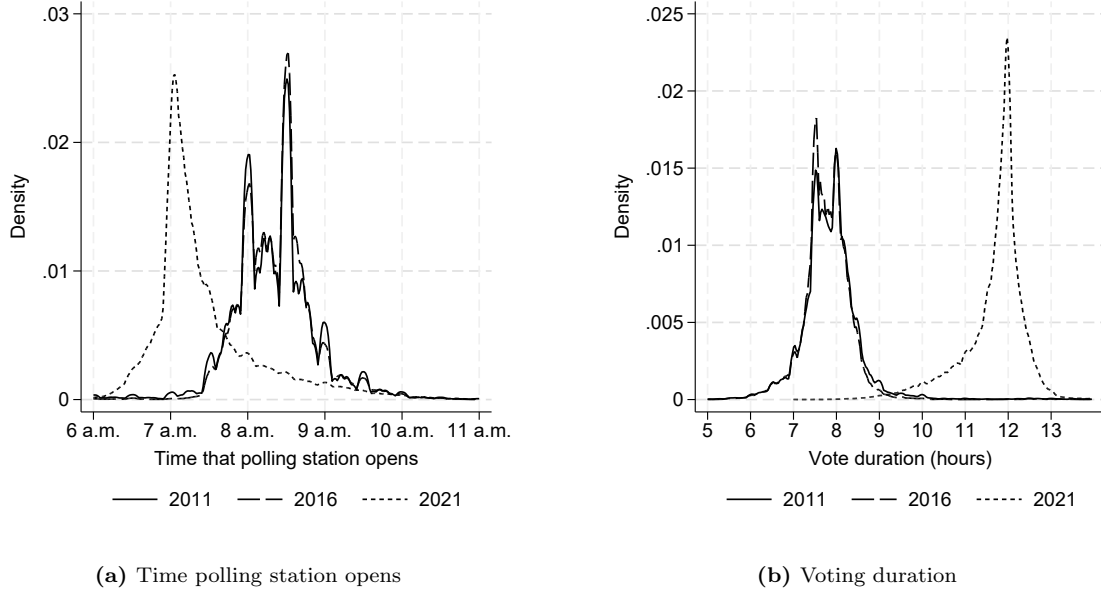
Notes: Data from the World Values Survey (2017–2022). The x-axis shows the share of surveyed individuals who responded “Not at all often” or “Not often” to the question “How often in your country’s elections: Election officials are fair?” The y-axis shows the share of surveyed individuals who responded “Not at all often” or “Not often” to the question “How often in your country’s elections: Votes are counted fairly?”.

Figure B.3: Research Design



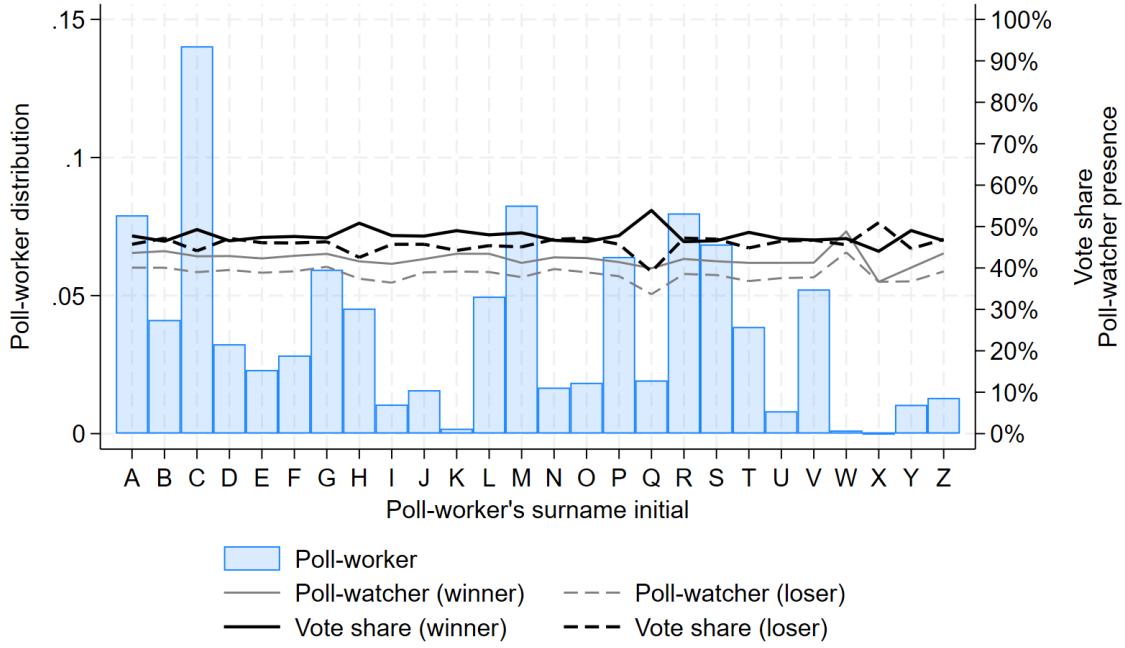
Notes: The following figures shows an example of an election site with 8 tables. Each of these has one of 4 possible observer assignments (red and blue). The analysis focuses on how each of the 4 possible cases deviates from the site mean considering that once controlling for first round votes, within a site, there is not much more than parties can do to inform their allocation that would be related to potential outcomes.

Figure B.4: Distribution of polling station opening times, vote duration, and vote count duration.



Notes: Panel (a) shows the distribution of polling station opening times across election years. Official opening time is 8:00 a.m. in 2011 and 2016, and 7:00 a.m. in 2021. In 2011, 29.4% of stations opened on time, with 92.3% open within one hour and 99.2% within two hours. In 2016, 24.3% opened on time, 92.6% within one hour, and 99.5% within two hours. In 2021, 23.1% opened on time, 83.5% within one hour, and 95.2% within two hours. Panel (b) shows the distribution of total voting duration. Polls were scheduled to close at 4:00 p.m. in 2011 and 2016, and at 7:00 p.m. in 2021; the extended closing time in 2021, introduced due to the COVID-19 pandemic, explains the longer voting durations observed that year. Panel (c) shows the distribution of vote-counting duration.

Figure B.5: Validating random assignment



Notes: The left axis shows the distribution of poll workers and poll-watchers by first initial of the lead poll worker's last name. The right axis shows the average vote share for the winner and runner-up parties. The x-axis shows the first initial of the lead poll worker's last name, which serves as a proxy for the last names of voters assigned to that classroom, since voters are assigned to classrooms alphabetically. The figure shows no strong association between alphabetical assignment and voting patterns, which supports the assumption that classrooms can be considered similar within a school. Moreover, since these are national averages, within-site variation would be even smaller. Main estimates are robust to excluding initials that exhibit the largest differences in vote shares.